

CO2-AT1-Part2

Application-Based Questions

Digital Forensics using TD (0), SARSA & Q-Learning

1. Using Q-Learning

1. Pseudocode

- Initialize $Q(s, a) = 0$ for all states and actions
- Set learning rate α , discount factor γ , exploration rate ϵ
- For each episode do
- Initialize state s
- Repeat until terminal state
- Choose action a using ϵ -greedy policy
- Execute action a
- Observe reward r and next states'
- Update Q-value
- $s = s'$
- End Repeat
- End For

Return optimal Q-table and forensic decision policy

2. Algorithm with Step-by-Step Formula

Step 1: Initialize Environment

Define

- States: $S = \{s_1, s_2, \dots, s_n\}$
- Actions: $A = \{a_1, a_2, \dots, a_m\}$
- Initialize

$$Q(s, a) = 0 \quad (1)$$

Eq. (1): Initializes the Q-table with zero values for every state-action pair before learning begins.

Step 2: Select Action

Choose an action using the ε -greedy strategy.

$$a = \begin{cases} \text{Random Action,} & \text{with probability } \varepsilon \\ \arg \max_a Q(s, a), & \text{otherwise} \end{cases} \quad (2)$$

Eq. (2): Selects either a random action for exploration or the action with the highest Q-value for exploitation.

Step 3: Compute Reward

After examining the digital evidence,

$$R = \begin{cases} +10, & \text{Correct evidence identified} \\ -5, & \text{Incorrect evidence identified} \end{cases} \quad (3)$$

Eq. (3): Assigns positive rewards for correct forensic decisions and penalties for incorrect decisions.

Step 4: Update Q-Value

$$Q(s, a) = Q(s, a) + \alpha \left[R + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (4)$$

Eq. (4): Updates the Q-value using the reward received and the estimated future reward, improving the decision policy over time.

Step 5: Prediction

$$\pi^*(s) = \arg \max_a Q(s, a) \quad (5)$$

Eq. (5): Selects the optimal action with the highest learned Q-value to predict the best forensic investigation step.

3. Calculation and Prediction

Given

- Learning rate

$$\alpha = 0.5$$

- Discount factor

$$\gamma = 0.9$$

- Current Q-value

$$Q(s, a) = 5$$

- Reward

$$R = 10$$

- Maximum future Q-value

$$\max Q(s', a') = 8$$

Calculation

Using Eq. (4)

$$\begin{aligned} Q(s, a) &= 5 + 0.5[10 + 0.9(8) - 5] \\ &= 5 + 0.5[10 + 7.2 - 5] \\ &= 5 + 0.5(12.2) \\ &= 5 + 6.1 \\ Q(s, a) &= 11.1 \end{aligned}$$

Prediction

Since the updated Q-value is **11.1**, the agent predicts that this forensic investigation action is optimal. During future digital forensic investigations, the system will prioritize this action because it has learned that it produces the highest expected reward for identifying valid digital evidence.

2. Digital Forensics using SARSA

1. Pseudocode

- Initialize $Q(s,a)=0$ for all states and actions
- Set learning rate α , discount factor γ , exploration rate ϵ
- For each episode
- Initialize state s
- Choose action a using ϵ -greedy policy
- Repeat until terminal state
- Take action a
- Observe reward r and next state s'
- Choose next action a' using ϵ -greedy
- Update $Q(s,a)$
- $s = s'$
- $a = a'$
- End Repeat
- End For
- Return optimal forensic investigation policy

2. Algorithm with Step-by-Step Formula

Step 1: Initialize Q-Table

$$Q(s, a) = 0 \quad (1)$$

Eq. (1): Initializes the Q-table with zero values for all state-action pairs before the learning process begins.

Step 2: Select an Action

Choose the current action using the ϵ -greedy policy.

$$a = \begin{cases} \text{Random Action,} & \epsilon \\ \arg \max_a Q(s, a), & 1 - \epsilon \end{cases} \quad (2)$$

Eq. (2): Balances exploration and exploitation while selecting the next forensic investigation action.

Step 3: Receive Reward

$$R = \begin{cases} +8, & \text{Evidence correctly identified} \\ -4, & \text{False evidence detected} \end{cases} \quad (3)$$

Eq. (3): Rewards correct forensic evidence detection and penalizes incorrect investigation outcomes.

Step 4: Update the Q-Value

$$Q(s', a) = Q(s, a) + \alpha[R + \gamma Q(s', a') - Q(s, a)] \quad (4)$$

Eq. (4): Updates the Q-value using the current reward and the Q-value of the next selected action, allowing the agent to learn an improved investigation policy.

Step 5: Prediction

$$\pi(s) = \arg \max_a Q(s, a) \quad (5)$$

Eq. (5): Predicts the best forensic action by selecting the action with the highest learned Q-value.

3. Calculation and Prediction

Given

- Learning rate

$$\alpha = 0.4$$

- Discount factor

$$\gamma = 0.9$$

- Current Q-value

$$Q(s, a) = 6$$

- Reward

$$R = 8$$

- Next state-action value

$$Q(s', a') = 7$$

Calculation

Using Eq. (4)

$$\begin{aligned} Q(s, a) &= 6 + 0.4[8 + 0.9(7) - 6] \\ &= 6 + 0.4[8 + 6.3 - 6] \\ &= 6 + 0.4(8.3) \\ &= 6 + 3.32 \\ Q(s, a) &= 9.32 \end{aligned}$$

Prediction

The updated **Q-value = 9.32**, indicating that the selected investigation action is effective. During future digital forensic investigations, the SARSA agent will prefer this action because it has learned that it leads to a higher expected reward for accurate evidence identification.

3. Digital Forensics using TD (0)

1. Pseudocode

- Initialize $V(s)=0$ for all states
- Set learning rate α and discount factor γ
- For each episode
- Initialize state s
- Repeat until terminal state
- Take an action
- Observe reward R and next state s'
- Update state value $V(s)$
- $s = s'$
- End Repeat
- End For
- Return optimal state-value function

2. Algorithm with Step-by-Step Formula

Step 1: Initialize State Values

$$V(s) = 0 \quad (1)$$

Eq. (1): Initializes the value function for every forensic investigation state to zero before learning begins.

Step 2: Observe Current State and Reward

The investigator moves from the current state to the next state after examining digital evidence.

$$(s, R, s') \quad (2)$$

Eq. (2): Represents the transition from the current forensic state to the next state while receiving an immediate reward.

Step 3: Compute Temporal Difference Error

$$\delta = R + \gamma V(s') - V(s) \quad (3)$$

Eq. (3): Computes the temporal difference error by comparing the current state value with the estimated value of the next state.

Step 4: Update the State Value

$$V(s) = V(s) + \alpha \delta \quad (4)$$

Eq. (4): Updates the state value using the temporal difference error to improve future forensic decision-making.

Step 5: Prediction

$$\hat{V}(s) = \max V(s) \quad (5)$$

Eq. (5): Predicts the most valuable forensic investigation state by selecting the state with the highest learned value.

3. Calculation and Prediction

Given

- Learning rate

$$\alpha = 0.3$$

- Discount factor

$$\gamma = 0.9$$

- Current state value

$$V(s) = 5$$

- Reward

$$R = 8$$

- Next state value

$$V(s') = 7$$

Calculation

Using Eq. (3),

$$\begin{aligned}\delta &= 8 + 0.9(7) - 5 \\ &= 8 + 6.3 - 5 \\ &= 9.3\end{aligned}$$

Using Eq. (4),

$$\begin{aligned}V(s) &= 5 + 0.3(9.3) \\ &= 5 + 2.79 \\ V(s) &= 7.79\end{aligned}$$

Prediction

The updated **state value is 7.79**, indicating that the current forensic investigation state is highly beneficial. The TD(0) agent predicts that similar investigation paths should be followed in future cases because they are expected to produce better evidence collection and analysis outcomes.